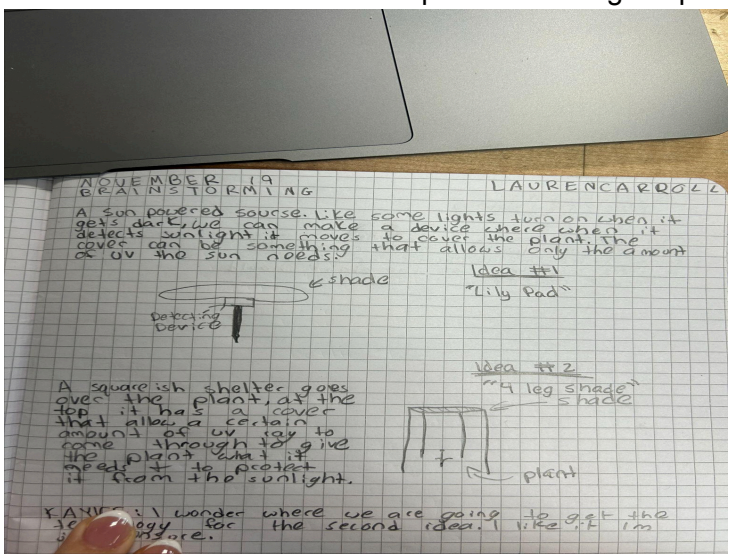
Element D Initial Template

Element D: Design Concept Generation, Analysis, and Selection

Design Idea #1 Lauren Carroll Date: November 21
Description: A sun powered source. When certain lights come on when it gets dark, this solar source detects sunlight and shifts over the plant so the plant is not in direct sunlight. In the top of the source it has holes that allow water to come through when it rains. As it gets dark, the shade shifts back over off of the plant and charges up for the next day.

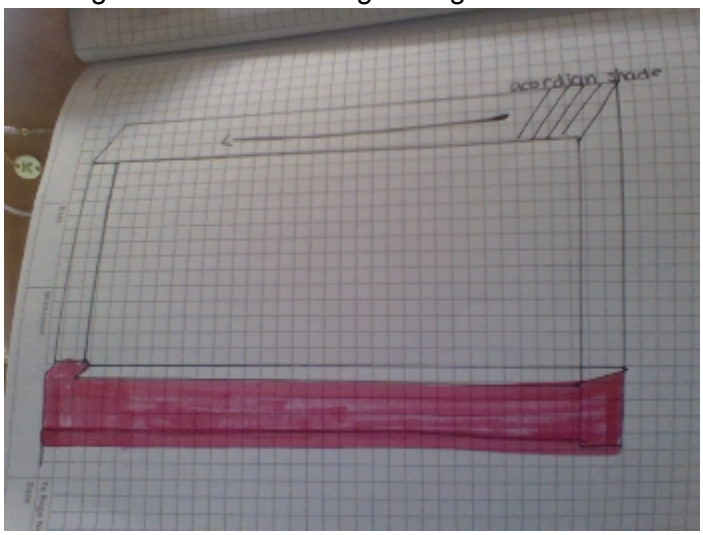


Design Idea #2

Kaylee Vaughn

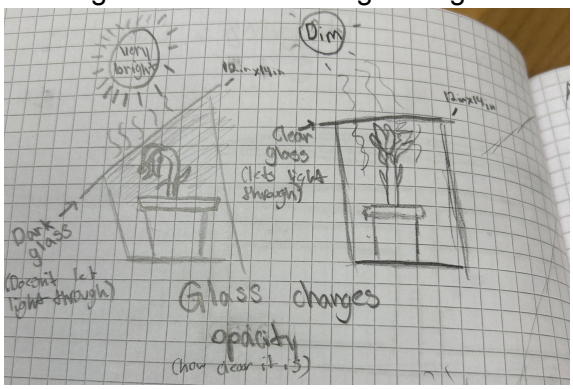
Date:11/21
Description: This device will be set on a timer and the accordion shade will move out on “wheels” or tracks. You will set and timer for a certain time of day and the shade will move out on a schedule. Depending on how much sun the plant needs everyday. It uses solar panels to charge so you don’t have to change it.

Drawing that includes distinguishing information such as scale, dimensions, and materials:



Design Idea #3 Kylen Roach

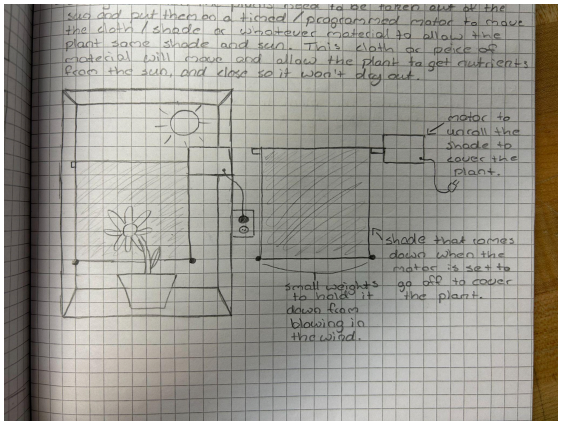
Date: November 21
Description: A device that uses a special type of glass that under certain amounts of sunlight changes its shade and turns into tinted glass that blocks sunlight.
Drawing that includes distinguishing information such as scale, dimensions, and materials:



Design Idea #4 Ethan Englebright

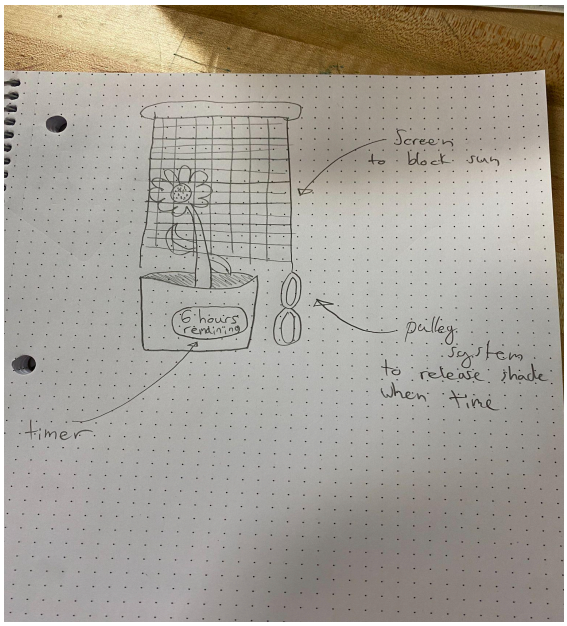
6

Date: November 21, 2024
Description: A device that automatically moves back and forth to provide shade over the plant we are given. This device would be on a timer that controls a motor; this motor moves a cloth with weights on the bottom to hold it down.
Drawing that includes distinguishing information such as scale, dimensions, and materials:



Design Idea #5 Carson Reece

Date: November 21, 2024
Description: A device that utilizes a shading cover that drops down to shade the plant on a timer using a pulley system.
Drawing that includes distinguishing information such as scale, dimensions, and materials:



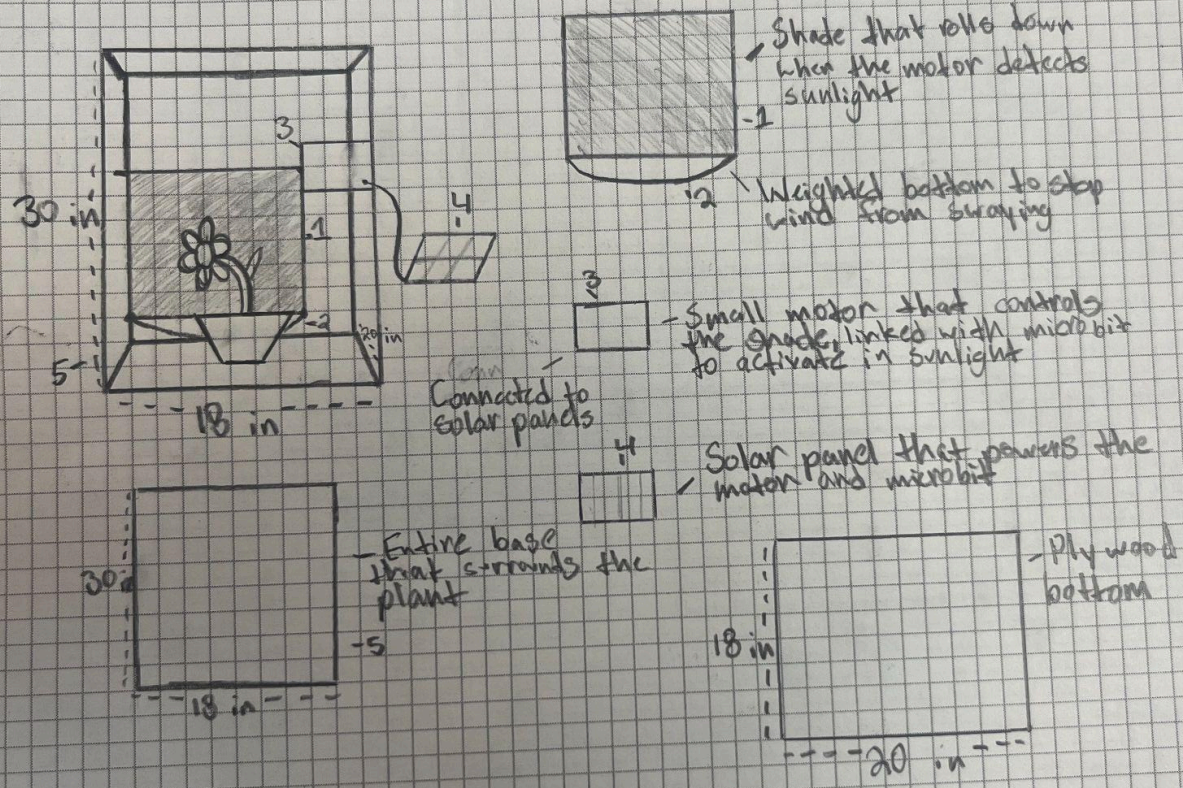
	weight	Option 1	Option 2	Option 3	Option 4	Option 5
Moves based on sunlight	40%	4 1.6	4 1.6	4 1.6	4 1.6	3 1.2
Programmable for time of day and total time in the sun.	30%	4 1.2	4 1.2	4 1.2	5 1.5	1 .3
Must use a battery	10%	1 .1	3 .3	3 .3	3 .3	1 .1
The device must last up to a week	10%	2 .2	5 .5	5 .5	5 .5	3 .3
Accommodate a seedling tray	5%	1 .05	5 .25	4 .2	5 .25	3 .15
Hold up to 10 lbs.	5%	1 .05	2 .1	3 .15	2 .1	3 .15
score	100%	3.2	4	3.95	4.25	2.2

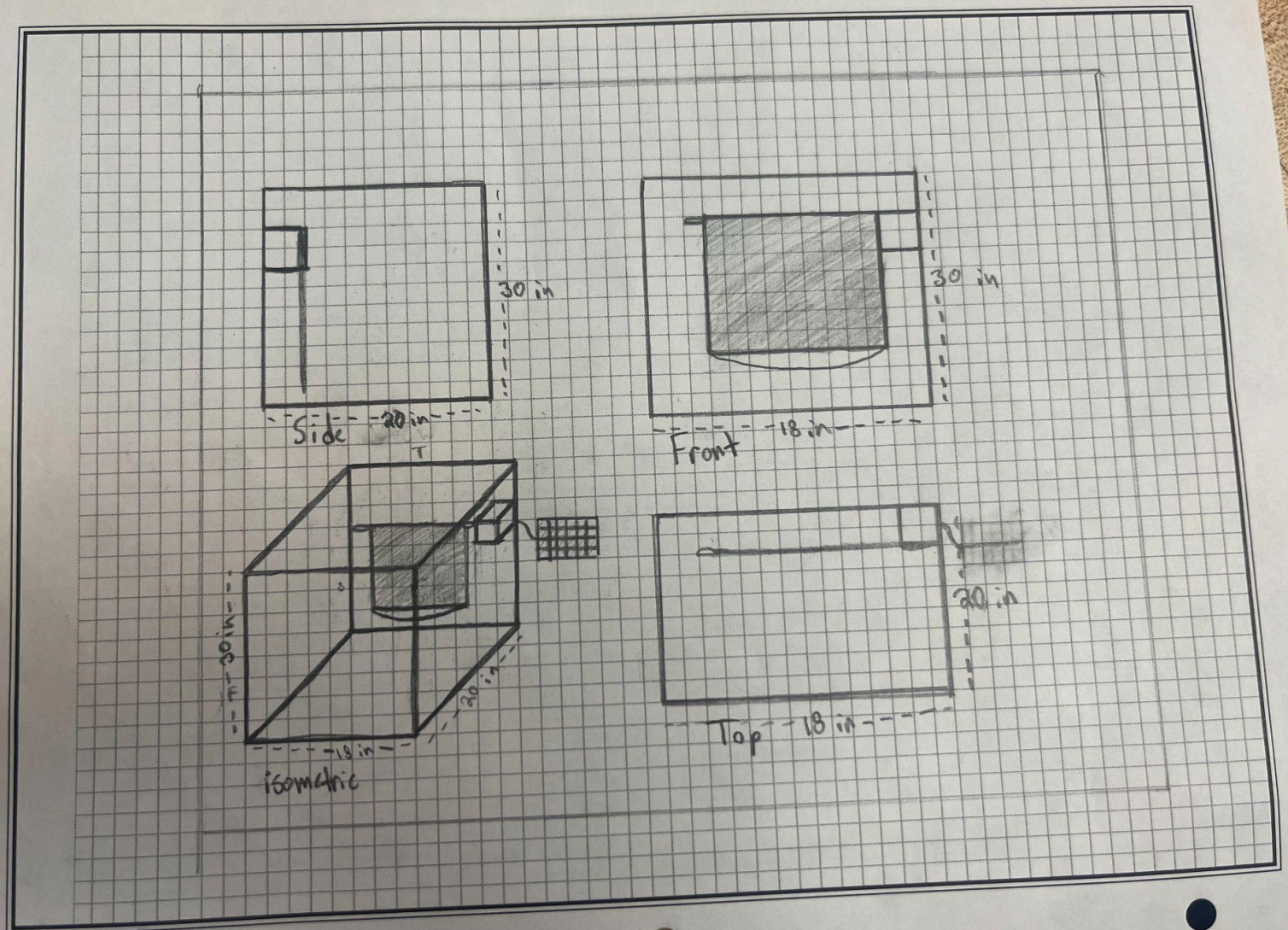
Design Blueprint (D4)

Design Blueprint #1

Date:12/6

Sketch: Final Design


30 in
18 in
20 in
5
1
2
3
4
Shade that rolls down when the motor detects sunlight
Weighted bottom to stop wind from swaying
Small motor that controls the shade, linked with micro bit to activate in sunlight
Connected to solar panels
Solar panel that powers the motor and microbit
Entire base that surrounds the plant
Plywood bottom


30 in
Side
20 in
30 in
Front
18 in
30 in
18 in
20 in
isometric
20 in
Top
18 in